MILITARY SPECIFICATION

AIR-CONDITIONING COMPONENT: FAN-COIL ROOM UNITS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 200 to 600 cubic feet per minute (cfm) factory-assembled, room air-conditioning (heating and cooling) units for remote installation. The sources of heating and cooling are not included.

1.2 Classification. The air-conditioning units shall be of the following types, styles, and sizes as specified (see 6.2):

Type I - Single coil.
Type II - Dual coil.

Style A - Vertical air flow.
Style B - Horizontal air flow.

Size 200 - 200 cfm (capacity).
Size 300 - 300 cfm (capacity).
Size 400 - 400 cfm (capacity).
Size 600 - 600 cfm (capacity).

1.2.1 Definitive military specification part number. The definitive military specification part number for items described in this specification will be identified as shown in 6.5.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4130

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SPECIFICATION

MILITARY

MIL-R-3593 - Refrigeration and Cooling Equipment (Excluding Household Refrigerators) Packaging of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

ARI 440 - Standard for Room Fan-Coil Air-Conditioners.

(Application for copies should be made to the Air-Conditioning and Refrigeration Institute, 1815 N. Fort Myer Drive, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B1.1 - Unified Inch Screw Threads (UN and UNR Thread Form).
B2.1 - Pipe Threads (Except Dry Seal).
B2.2 - Dry Seal Pipe Threads.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNDERWRITERS' LABORATORIES, INC. (UL)

Recognized Components Index.

UL 883 - Fan-Coil Units and Room Fan-Heater Units.
UL 900 - Test Performance of Air Filter Units.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. Each unit shall consist essentially of a water-coil assembly or assemblies, drain pan(s), outside air damper when required, air filter(s), fan(s), motor(s), fan drive, motor switch, and enclosure. Devices or means integral with the unit shall be provided for leveling vertical units. The unit shall comply with ARI Standard 440 and UL 883.

3.1.1 Type I. The type I unit shall have a single coil to be used with chilled water for cooling and with hot water for heating.

3.1.2 Type II. The type II unit shall be designed for use with a four-pipe system, with one coil for chilled water and a separate coil for hot water, or a single coil with separate circuits for chilled and hot water.

3.2 First article. When specified (see 6.2), the contractor shall furnish a fan-coil room unit of the type, style, and capacity as required for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Certification. Prior to approval of the sample fan-coil unit furnished for first article inspection, if one is required, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the fan-coil unit he proposes to supply under this specification meets the requirements of ARI Standard 440 and UL 883. Acceptable evidence of meeting these standards will be as follows:

a. ARI Standard - The label, listing, and data showing compliance with ARI 440.
b. UL Standard - The Underwriters' Label required for Fan-Coil Units and Room Fan-Heater Units conforming to UL 883.

3.3.1 Alternate certification. In lieu of the certification required under 3.3, acceptable evidence of meeting the requirements of the applicable standards will be a certified test report from an independent testing laboratory, selected by the manufacturer and approved by the contracting officer, indicating that a unit identical to the units to be furnished under this specification has been examined and tested, in accordance with the respective ARI Standard 440 and UL 883 and meets the requirements thereof. When authorized in writing by the contracting officer, such examination and tests for alternate certification may be conducted by the manufacturer at the manufacturer's plant when the cognizant inspection authority has determined that adequate facilities and instrumentation are available for conducting the required tests.

3.4 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

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3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Design. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. Drainage shall be in accordance with standard engineering practice. Unless otherwise specified (see 6.2), the equipment shall be designed for operation from a 120-volt, single-phase, 60-Hertz, alternating current power source. Screw threads for all thread securing or attaching devices and threaded parts shall be in conformance with ANSI B1.1, B2.1, or B2.2 as applicable.

3.8 Component parts.

3.8.1 Enclosure. The fan-coil units shall be recessed or semi-recessed, floor- (free standing), ceiling- or wall-mounted, as specified (see 6.2). The enclosure shall be constructed of steel, unless otherwise specified, and shall be properly reinforced and braced, and finished as here- inafter specified (see 6.2). The front panel shall be constructed of not lighter than 18 gage (0.0478 inch in nominal thickness) steel, shall be removable, and shall be provided with minimum 1/2-inch thick neoprene- or vinyl-coated glass-fiber insulation when required to prevent condensation. The discharge grilles shall be adjustable or nonadjustable type and shall be of such design as to properly distribute air throughout the conditioned space. Access doors shall be provided for all piping and control compartments. Plastic discharge or return grilles are acceptable.

3.8.2 Fans. Fans and scrolls shall be galvanized steel or aluminum. Fans shall be multibladed, centrifugal type. All surfaces shall be smooth. Assemblies shall be accessible for maintenance. Disassembly and reassembly shall be by means of mechanical fastening devices and not by epoxies or cements. All fans shall be balanced dynamically and statically.

3.8.3 Coils. Water coils shall be constructed of not less than 3/8-inch outside diameter seamless copper tubing with copper or aluminum fins mechanically bonded, brazed or soldered to the tubes. The coils shall be provided with not less than 1/2-inch outside diameter flare or sweat connectors and with manual air vents. The coils shall be suitable for 200 pounds per square inch gage (psig) working pressure and, when tested as specified in 4.6.1, shall show no leakage.

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3.8.4 Drain pans. Drain pan(s) shall be properly sized and located to collect all water condensed and dripping. The pans shall be either of galvanized steel not lighter than 18 gage (0.0516-inch nominal thickness); or from a single sheet of uncoated steel not lighter than 16 gage (0.0598-inch nominal thickness) and coated inside and outside with bitumastic material. All pans shall be covered with a sufficient thickness of insulation to prevent condensation, coated with a waterproofing material if the insulation is not nonhygroscopic. Drain pans shall be properly pitched to insure condensate runoff. Not less than 3/4-inch National Pipe Thread or 5/8-inch outside diameter copper drain connection shall be provided in drain pans.

3.8.5 Air damper. When specified (see 6.2), outside air damper shall be provided. Method of control shall be manual adjustment at the damper remote control, at the control junction box, or automatic damper motor and linkage for electric or pneumatic operators, all factory mounted, as specified (see 6.2). Means shall be provided in the outside intake-return air compartment to prevent outside air from passing through the return air grille directly into the room when the unit fan is off.

3.8.6 Filters. Filters shall be provided for each unit and shall be throw-away type or permanent washable type, as specified (see 6.2), 1/2-inch or 1-inch nominal thickness standard with the fan-coil unit manufacturer. The filters shall be class 1 or 2 in conformance with UL 900. Filters shall be removable with or without the use of simple hand tools.

3.8.7 Motors. Motors shall be split capacitor, two- or three-speed, impedance or overload-protected type, and directly connected to unit fan(s). Motor switch shall be two- or three-speed and off, manually operated, and shall be mounted on a properly identified plate inside the unit below or behind an access door. For a fully concealed unit the motor controller shall be provided with appropriate hardware for remote mounting. In lieu of the above fan-speed control, a solid-state variable speed controller having a minimum speed reduction of 50 $\pm$5 percent may be provided.

3.8.8 Wiring. All wiring required to connect electrical components and switches shall be provided.

3.9 Treatment and painting. Unless otherwise specified (see 6.2), the units shall be treated and painted in accordance with the manufacturer's standard practice. All surfaces of the units, other than corrosion-resisting steel, shall be protected against corrosion.

3.10 Lubrication. All moving parts of the unit normally requiring lubrication shall have means provided for such lubrication and shall be properly lubricated as specified in section 5

3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

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3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1 and 6.4).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one fan-coil unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the contractor's current inventory provided the fan-coil unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining fan-coil units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the samples selected in accordance with 4.4. This inspection shall include the examination of 4.5, the tests of 4.6, and the requirements of 4.7.

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4.3 Inspection lot. All units of the same type, style, and capacity offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.4 Sampling. A random sample of fan-coil units shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. The inspection level shall be II with an Acceptable Quality Level (AQL) of 1.5 percent defective.

4.4.2 Sampling for tests. The inspection level shall be S-3 with an AQL of 4.0 defects per 100 units.

4.4.3 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.5 Examination. The fan-coil units shall be visually inspected to determine conformance to the requirements of this specification. Examinations shall be made on each sample selected in accordance with 4.4.1 and shall encompass all visual examinations and measurements as required to prove the workmanship, reveal errors and omissions of the production process, hidden defects, unauthorized substitution of materials, deviations from design, and other factors which may be deemed necessary to verify conformance with the requirements of this specification. Any fan-coil unit in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6 Tests. Each sample selected in accordance with 4.4.2 shall be tested as specified. Failure of any sample unit to pass any test shall be cause for rejection, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.6.1 Leakage test. The coils shall be hydrostatically tested at 250 psig for not less than 1-minute duration or under water at 300-psig air pressure for not less than 10-seconds duration.

4.7 Packaging inspection. The sampling and inspection of the preservation, packaging, packing, and marking of the fan-coil units shall be in accordance with the requirements of section 4 of MIL-R-3593. The inspection shall consist of the quality conformance inspection of MIL-R-3593. A preproduction pack shall be furnished for examination and test when required (see 6.2), and within the time frame specified (see 6.2), in accordance with MIL-R-3593.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-R-3593 with the level of preservation and level of packing as specified (see 6.2).

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6. NOTES

6.1 Intended use. A room fan-coil unit is a factory-made assembly which provides the functions of circulation, cooling, or cooling and heating, and filtering of air but which does not include the source of cooling or heating. A fan-coil unit is normally designed for free delivery of air into a room, but may be applied with minimal duct work having a static resistance not exceeding 1/4 inch of water. Fan-coil unit may be designed for furred-in application or with an enclosure for application within the conditioned space.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Type, style, and capacity required (see 1.2).
c. When a first article is required (see 3.2).
d. Electrical characteristics, if other than specified (see 3.7).
e. Type of enclosure and mounting (see 3.8.1).
f. Material for enclosure if other than steel (see 3.8.1).
g. Outside air damper and manual or automatic operators (see 3.8.5).
h. Whether throw-away or washable filters are required (see 3.8.6).
i. Treatment and painting if other than manufacturers' standard practice (see 3.9).
j. When a preproduction pack is required (see 4.7).
k. Time frame required for submission of the preproduction pack (see 4.7).
l. Level of preservation, packing, marking, and level of packing required (see 5.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete fan-coil unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitive military specification part number. The definitive military specification part number corresponds to the type, class, and other specific classification data applicable to fan-coil units covered by this specification, and defines the requirements of the options presented under this specification. The military specification number, the type and class code number, and the fan-coil units' other applicable code numbers are combined to form the definitive military specification part number.

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6.5.1 Cataloging data. For cataloging purposes, part numbers for the fan-coil units are assigned as follows:

A-52670A X X X X

Military Specification Number ----------
Type and Style Code Number ----------
Size and Enclosure Model ----------
Materials of Enclosures ----------
Damper Types and Filter Styles ----------

6.5.2 Type and style. The type and style of fan-coil units are identified by a single number (see table I).

TABLE I. Code number to type and style.

	Type I	Type II
Style A	1	2
Style B	3	4

6.5.3 Size and enclosure model. The size (cfm) and model of the fan-coil units are identified by a single alphabetic character (see table II).

TABLE II. Code letter to size and model.

	Size			
Model	200 cfm	300 cfm	400 cfm	600 cfm
Recessed	a	b	c	d
Semi-recessed	e	f	g	h
Floor-mounted (free standing)	i	j	k	l
Ceiling-mounted	m	n	p	r
Wall-mounted	s	t	u	v

6.5.4 Enclosures. The materials used for the enclosures are identified by a single number (see table III).

TABLE III. Materials for enclosures.

	Steel galvanized	Aluminum
Enclosures	7	8

6.5.5 Dampers and filters. The type of dampers (manual or automatic) and the style of air filters (permanent or disposable) are identified by a single alphabetic character (see table IV).

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TABLE IV. Dampers and filters.

	Dampers	
	Manual	Automatic
Permanent	A	B
Disposable	C	D

6.5.6 Motor controls and painting options. The type of motor controls and painting options are identified by a single number (see table V).

TABLE V. Motor controls and painting options.

Motor Control	Painting	
	Manufacturer's standard	(as specified in 6.2)
Switch (2- or 3-speed and off position)	1	2
Variable solid-state (100 percent to 50 percent and off position)	3	4

6.6 Changes from previous issue. Asterisks are not used in this revision to identify the changes with respect to the last previous issue due to the extensiveness of the changes involved.

Custodians:

Army - ME
Navy - YD

Preparing activity:

Navy - YD

Project No. 4130-0269

User activity:

Army - CE

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-A-52670A - AIR-CONDITIONING COMPONENT: FAN-COIL ROOM UNITS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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Replaces edition of 1 Jan 72 which may be used.

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